AQ9.4: Activity Questions 4 - Not Graded

This assignment will not be graded and is only for practice.
Limits:

Level 1:

Find the value of the following limit:

$$\lim_{(x,y)\to(2,3)} \frac{x^2+y^2-y}{y^2-x^2}$$ lim(x,y)→(2,3) y2−x2x2+y2−y

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 2
1 point
1 point
Suppose f f and g g are multivariable functions defined on domain R^2 R2 with the following limits:

$$\lim_{(x,y)\to(0,0)} f(x,y) = l \text{ and} \lim_{(x,y)\to(0,0)} g(x,y) = k$$ lim(x,y)→(0,0) f(x, y) = l and lim(x,y)→(0,0) g(x, y) = k

where $l, k \in R$ l, k ∈ R. Which of the following options is(are) correct?

- [ ] $\lim_{(x,y)\to(0,0)} (f(x,y)+g(x,y)) = l+k$ (x,y)→(0,0)lim (f(x, y) + g(x, y)) = l + k
- [ ] $\lim_{(x,y)\to(0,0)} (f(x,y).g(x,y)) = lk$ (x,y)→(0,0)lim (f(x, y).g(x, y)) = lk
- [ ] $\lim_{(x,y)\to(0,0)} (5f(x,y)) \neq 5l$ (x,y)→(0,0)lim (5f(x, y)) = 5l
- [ ] $\lim_{(x,y)\to(0,0)} \left(\frac{f(x,y)}{g(x,y)}\right) = \frac{l}{k}$ (x,y)→(0,0)lim (g(x,y)f(x,y)) = kl, when $k \neq 0$ k = 0

No, the answer is incorrect.
Score: 0
Accepted Answers:
$\lim_{(x,y)\to(0,0)} (f(x,y)+g(x,y)) = l+k$ (x,y)→(0,0)lim (f(x, y) + g(x, y)) = l + k

$\lim_{(x,y)\to(0,0)} (f(x,y).g(x,y)) = lk$ (x,y)→(0,0)lim (f(x, y).g(x, y)) = lk

$\lim_{(x,y)\to(0,0)} \left(\frac{f(x,y)}{g(x,y)}\right) = \frac{l}{k}$ (x,y)→(0,0)lim (g(x,y)f(x,y)) = kl, when $k \neq 0$ k = 0

Consider the function $f: R^2 \longrightarrow R$ f : R2 ⟶ R defined as: $f(x,y) = \begin{cases} \frac{4x^2y^2}{x^2+y^2} & x, y \neq 0 \\ 0 & x = y = 0 \end{cases}$ f(x, y) = {x2+y24x2y20x, y = 0x = y = 0What is the value of $\lim_{(x,y)\to(0,0)} (f(x,y))$(x,y)→(0,0)lim (f(x, y))?

No, the answer is incorrect.
Score: 0

Accepted Answers:
(Type: Numeric) 0
1 point
Suppose f and g are multivariable functions defined on domain R^2 as
$f(x,y) = \frac{x^2+y^2+y(y-2x-1)}{x^2-y^2}$ and $g(x,y) = \frac{2y^2+x(x-2y-1)}{x^2-y^2}$. Find the value of $\lim_{(x,y)\to(1,1)} \left(f(x,y)-g(x,y)\right)$.

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 0.5
1 point
Find the value of the following limit:

$$\lim_{(x,y)\to(3,1)} \frac{x^2-3xy^2}{x^2-9y^4}$$

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 0.5
1 point

Level 2:

$\because$ Consider the function $f: R^2 \longrightarrow R$ defined as:

$$f(x,y) = \begin{cases} \frac{3xy}{x^2+y^2} & x, y \neq 0 \\ 0 & x = y = 0 \end{cases}$$

Use the given information to answer the questions 6-8:

If f approaches to L as (x,y) approaches to the origin along Y-axis, then find the value of L?
No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 0
1 point
If f approaches to L as (x,y) approaches to the origin along a straight line $y = 3x$, then find the value of L
No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 0.9
1 point

1 point
Which of the following options is(are) false?
☐

$\lim_{(x,y)\to(0,0)} f(x,y) = 0$

☐ $\lim_{(x,y)\to(0,0)} f(x,y) = 1$

☐ Limit of the given function at (0,0) exists but is not equal to 0.
☐ Limit of the given function at (0,0) does not exist.

No, the answer is incorrect.
Score: 0
Accepted Answers:

$\lim_{(x,y)\to(0,0)} f(x,y) = 0$

$\lim_{(x,y)\to(0,0)} f(x,y) = 1$

Limit of the given function at (0,0) exists but is not equal to 0.

1 point

Which of the following functions have limit value equal to 5 at the point (1,1,-1)?

☐ $f(x,y,z) = \frac{x+3yz^2-zx}{xy+yz+xz^2}$

☐ $f(x,y,z) = \frac{3(x+y-z)+7x^2y+yz}{xyz+zy+xz}$

☐ $f(x,y,z) = x^3+y^3+z^3-2xy-2yz-4zx$

☐ $f(x,y,z) = \frac{x^2e^{\frac{x}{2}}+y^2e^{\frac{y}{2}}+z^2e^{\frac{-z}{2}}}{e^{\frac{x+y+z}{2}}}$

No, the answer is incorrect.
Score: 0
Accepted Answers:

$f(x,y,z) = \frac{x+3yz^2-zx}{xy+yz+xz^2}$

$f(x,y,z) = x^3+y^3+z^3-2xy-2yz-4zx$

1 point

Consider the function $f: R^2 \longrightarrow R$ defined as:

$$f(x,y) = \begin{cases} \frac{x^k y}{x^{2n}+y^{2n}} & x, y \neq 0 \\ 0 & x = y = 0 \end{cases}$$

where $k, n \in N \setminus \{0\}$. Which of the following statements is(are) true about f?

☐ If $k = 2n-1$, then the limit at (0,0) exists and is equal to 0.

☐ If $k < 2n-1$, then the limit at (0,0) does not exist.

☐ If $k > 2n$, then the limit at (0,0) always exists and is equal to 0.

☐ If $k > 2n$, then the limit at (0,0) does not exist.

No, the answer is incorrect.
Score: 0
Accepted Answers:

If $k < 2n-1$, then the limit at (0,0) does not exist.
If $k > 2n$, then the limit at (0,0) always exists and is equal to 0.

Check Answers

Your score is: 0/10